PRIME FACTOR STUDIO — IN-BOX USER GUIDE

Kinesthetic Number Theory, Proportional Logarithmic Stacking & 12-Socket Lid Tray

CARD 05 / 10 • PRIME-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · 4·Fractions · **5·Primes** · 6·Algebra · 7·Probability · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

FEEL THE PRIMES. MULTIPLY WITH HEIGHT.

Standard base-10 counting blocks add lengths (1+1=2).
Prime Factor blocks are logarithmically proportional:
$\log(a \times b) = \log(a) + \log(b)$
Stacking blocks in a tower multiplies their numbers!

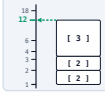
CALIBRATED LOGARITHMIC DATUM

- Unit Datum: Block [2] = 1U = 7.00 mm visible height
- Log Modulus: $K_{log} = 23.2535$ mm / decade
- Prime Formula: $h_{vis}(p) = K_{log} \times \log10(p)$
- Connector Spec: 1.85mm base plug & stacking socket

12-SOCKET LID TRAY & LOGARITHMIC SCALE

Flip 13U lid over to use 12-socket comparison stage.
Equal heights prove equal mathematical products!

VISUAL PROOF: TOWER 12 = [2] + [2] + [3]



▲ SUMMIT = 12
2 × 2 × 3 = 12
Visible: 25.09 mm

Commutative:
[3]+[2]+[2] = same height!

2. KIT PARTS INVENTORY (72 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Prime Block [2]	29x	7.00 mm (1.00U)	Blue
Prime Block [3]	17x	11.09 mm (1.58U)	Gold
Prime Block [5]	8x	16.25 mm (2.32U)	Classic White
Prime Block [7]	4x	19.65 mm (2.81U)	Slate Gray
Prime Block [11]	3x	24.21 mm (3.46U)	Gold
Prime Block [13]	3x	25.90 mm (3.70U)	Classic White
Prime Block [17]	2x	28.61 mm (4.09U)	Slate Gray
Prime Block [19]	2x	29.73 mm (4.25U)	Gold
Prime Block [23]	2x	31.67 mm (4.52U)	Slate Gray
13U Storage Bin	1x	91.0 mm (13U)	84.0mm Waterline
Stage Lid	1x	10.4 mm Stage	12-Socket Lid Tray

TOTAL: 70 Prime Blocks + Storage Bin + Comparison Stage Lid = 72 Parts

4-COLOR HIGH-CONTRAST PALETTE

- 4-Color Palette: Neighboring prime columns have distinct hues.
- Instant Sorting: Enables fast visual sorting across 70 blocks.
- High Durability: Scratch-resistant debossed numerals.
- Tactile Alignment: Connectors mate smoothly without binding.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **10-SECOND SELF-CHECKING PACK-UP**

Pack-up is a built-in math game! Each floor socket is labeled with its tower's product: 2^{12}=4096, 3^7·2^4=4374, 5^5=3125, 7^3·2^4=4802...
When all 70 blocks are stacked in their bays, all 12 towers level at the 84.0mm waterline. Stage lid latches firmly for quiet carry.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

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Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

FUNDAMENTAL THEOREM OF ARITHMETIC
Every integer > 1 has a unique prime factorization decomposition.
Universal 7.00mm Datum • Classroom Continuum (10 Kits)

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SERIES 1 • CARD 05 OF 10 • DOMAIN: FUNDAMENTAL THEOREM OF ARITHMETIC • LOGARITHMIC PROPORTIONAL TOWERS • FLUSH DATUM

LESSON GUIDE: HOW TO FIND THE LEAST COMMON MULTIPLE (LCM)

Step-by-Step Hands-On Tutorial: Finding LCM(12, 18) = 36 & The Law of Conservation of Primes

Grades 4–9 Math & Pre-Algebra • Hands-On Classroom & Home STEM Studio Series

CARD 05 / 10 • PRIME-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/prime-studio/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.6.NS.B.4 (LCM & GCD)
- CCSS.MATH.CONTENT.4.OA.B.4 (Factor Pairs)
- TEKS 111.26.B.6.7.A (Prime Factorization)

Grades 4–9 Math & Pre-Algebra

💡 **MATHEMATICAL INQUIRY**

Which prime blocks are shared by both towers, and which blocks are unique to one tower?

1 STEP 1: FACTOR & BUILD TOWERS 12 & 18

Decompose numbers 12 and 18 into prime building blocks:
- **Tower 12: Stack [2] + [2] + [3] (2 × 2 × 3 = 12)**
- **Tower 18: Stack [2] + [3] + [3] (2 × 3 × 3 = 18)**

Seat each tower in the 12-socket lid tray. Their summits **measure 25.09 mm (Tower 12) and 29.18 mm (Tower 18)!**

2 STEP 2: FIND SHARED OVERLAP (THE GCD)

Stand Tower 12 and Tower 18 side-by-side.
Identify the blocks they share in common (intersection):
- Tower 12 has: [2], [2], [3] | Tower 18 has: [2], [3], [3]

Shared Overlap: Exactly one [2] and one [3]!
Assemble GCD Tower: [2] + [3] = 6 (Summit: 18.09 mm).

3-4 STEPS 3 & 4: ASSEMBLE THE LCM TOWER

3. TAKE PRIME UNION: Take highest power of each prime: two [2]s, two [3]s.
4. STACK LCM TOWER: [2] + [2] + [3] + [3] = 36 (Summit: 36.19 mm)!
Calculator Verification: Press Factor Tree Icon ([PRIME] / [g][FACT]) and run 12 [ENTER] 18 [GCF] 12 [ENTER] 18 [F] [LCM] → 36.

5 CORE PRINCIPLE: CONSERVATION OF PRIMES: A × B = GCD × LCM

Multiply Tower 12 and Tower 18: you have three [2]s and three [3]s. Look at your GCD Tower ([2], [3]) and your LCM Tower ([2]², [3]¹): together they contain the EXACT same prime blocks!
Therefore: A × B = GCD(A, B) × LCM(A, B) = 6 × 36 = 216!

★ EXTENSION INVESTIGATION: FIND LCM(8, 12)

Build Tower 8: [2]+[2]+[2] and Tower 12: [2]+[2]+[3]. Prime union takes three [2]s and one [3]
→ LCM = 24 (28.09mm)! (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

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CROSS-REFERENCED CALCULATOR VERIFICATION
Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

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Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

PRIME-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 72 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
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3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California